

DBACT sampled discrete-dissipation: advisor delivery report

Static uniform-offset theorem_mode, $N = 16$, nine cases

Branch `feat/sampled-theorem-mode` on frozen baseline 98ba28e (not merged)

2026-09-11

Abstract

We repaired the sample-and-hold execution interface of the sampled `theorem_mode` controller and derived a discrete truncated-coverage inequality with a mass-weighted command-energy indicator J . This report delivers the honest verdict: the post-hoc numbers are informative and the conditional dissipation structure is sound, but there is *no* a priori centroid-error bound that turns the new estimate into a certified quantitative improvement over a model-free geometric ceiling, the $N = 16$ Hessian / local stability question remains open, and the static-oracle runs are theory baselines — **not** a certificate for unknown-object transport. All nine cases (L-shape, rectangle, C-channel $\times$ seeds 2/5/8) complete 30s with zero aborts; the C-channel seed 5 full-horizon J bound is explicitly *inapplicable*. The corrected derivation is in §A.

1 One-line verdict

Original coarse Theorem 1 bound: *no quantitative value* (it loses to a model-free geometry ceiling a priori). New discrete results: *conditional dissipation* plus *post-hoc explanatory value*. **No** a priori guarantee beating geometry. Static oracle $\neq$ unknown-object transport proved.

2 Point-by-point responses

A5 and hard switching. Assumption 5 (locally Lipschitz continuous-time feedback) is still incompatible with the endpoint-grid CVT path: hard disk / Voronoi membership on a finite 20×20 lattice produces finite centroid jumps under arbitrarily small site moves. The present branch does *not* claim the production CVT satisfies A5; it isolates a different interface: static sample-and-hold cover with wall rows, no position clipping, abort-on-illegal, and a Lipschitz-certified object-hold clearance.

Why the original Theorem 1 bound lacks quantitative value. Under the default static parameters the printed asymptotic formula evaluates to $\approx 1.90 \times 10^4$ for the gradient-squared residual, while a direct total-mass geometric ceiling is ≈ 8.99 , and measured 30s time averages are ≈ 0.011 – 0.013 . The formula is therefore worse than a model-free geometry bound by $\sim 2 \times 10^3$ and orders of magnitude above observation. Replacing the printed number by $\min(\text{formula}, \text{geometry})$ would hide the missing information, not create theory.

What the new discrete results establish.

1. **Proved (fixed ϕ^*).** The frozen truncated-partition majorant (Lemma A) gives the discrete inequality (1) without assuming dissipation.

2. **Proved** ($a = 2/k_c - \Delta > 0$). The exact-centroid / unsaturated pairing structure for a discrete decrease along centroid-aligned commands (Theorem B).
3. **Conditional**. Projection comparison against 0 holds only on steps in $\mathcal{K}_0$ (final QP set with the ρ margin); this is not promoted from “true on some L-runs” to a global fact — the C-channel seed 5 already exhibits six agent-steps with $0 \notin F_\rho$.
4. **Post-hoc**. With mass-weighted J_k, E_k , the finite-horizon form $(\star)$ matches L-shape measurements when $\bar{E}$ is taken from the trajectory.
5. **J is not the gradient residual**. Vanishing command speed does not imply vanishing $g^\star$.

$N = 16$ **Hessian**. No Hessian / curvature / invariant-neighborhood certificate was computed or claimed. This remains an open item, out of scope for this round by design.

A3 — same Gaussian kernel. The offset density and the continuous observer share the same unnormalized Gaussian edge kernel K_σ with a single σ and nonnegative weights; the manuscript should state this interface explicitly so Lemma-type constants have a clear domain.

3 Quantitative summary (static oracle, $N = 16$, 30 s)

Block	Theorem-1 / analytic	Geometric	Measured
$\ g^\star\ ^2$ avg	$\approx 1.90 \times 10^4$ (<i>not a cert.</i>)	≈ 8.99	$\approx 0.011\text{--}0.013$ (L)
Mass-weighted J	post-hoc $(\star) \approx 0.025\text{--}0.034$	$Mu_{\max}^2 \approx 0.18\text{--}0.23$ (preferred)	$\bar{J} \approx 0.009\text{--}0.012$
Centroid E	—	diameter $4R^2M \approx 4.80$ (loose)	$\bar{E} \sim 10^{-4}$ (<i>post-hoc</i>)

A priori error bound sufficient to beat geometry? No. Against the preferred geometric J ceiling $Mu_{\max}^2$, one needs $\bar{E} \leq E^\star$; the only model-only bound is the resolution-independent diameter bound $\approx 4.8 > E^\star$, which does not control grid/oracle centroid error. We refuse to treat trajectory $\max E$ as a uniform prior. Post-hoc, measured $\bar{E} \sim 10^{-4}$ makes $(\star)$ look useful — explanatory, not certified.

3.1 Nine-case table

All nine cases complete 30 s with zero aborts. The C-channel seed 5 records six zero-input-with- ρ failures (frames 241–246, agent 14) and is the only case with an *inapplicable* full-horizon J bound. The primary geometric J ceiling is $M_{\text{tot}}u_{\max}^2$ (*not* $Nm_+u_{\max}^2$).

shape	seed	full-horizon	H_0	$\bar{J}$	post-hoc J bd.	$Mu_{\max}^2$
l_shape	2	yes	0.9754	0.01108	0.03006	0.2295
l_shape	5	yes	0.9555	0.01071	0.02945	0.2295
l_shape	8	yes	1.0297	0.01152	0.03169	0.2295
rectangle	2	yes	0.8140	0.00894	0.02507	0.1803
rectangle	5	yes	0.8262	0.00910	0.02551	0.1803
rectangle	8	yes	0.8495	0.00930	0.02618	0.1803
c_shape	2	yes	1.0789	0.01201	0.03330	0.2295
c_shape	5	no	1.0884	0.01234 [†]	inapplicable	0.2295

shape	seed	full-horizon	H_0	$\bar{J}$	post-hoc J bd.	$Mu_{\max}^2$
c_shape	8	yes	1.1109	0.01251	0.03432	0.2295

[†] descriptive statistic tagged `not_covered_by_theorem`; the post-hoc bound is not applied because $\mathcal{K}_0 \neq [0, K)$. Exact machine values are in `nine_case_summary.json` and `nine_case_comparison.csv`; $Mu_{\max}^2$ values are recomputed offline from `observer.total_mass(vertices)`.

Three-command check (fixed). Each stage uses its own hypothetical next pose $P + \Delta u$; on refinement H , g and m are all taken from the same fine observer. Refined max slacks stay $O(10^{-6})$ – $O(10^{-7})$: numerical consistency at quadrature tolerance, **not** a strict numerical certificate (`no_rigorous_quadrature_error_bound`). Per-case values are in `three_command_hyp/`.

4 Figures

Discrete-average lines are the horizontal dashes; C-channel seed 5 condition failures are shaded. The discrete average $(1/K) \sum_k$ and the trapezoidal continuous-time approximation differ by $O(\Delta/T)$ (both reported in `j_bound/*.json`); the discrete mean is primary.

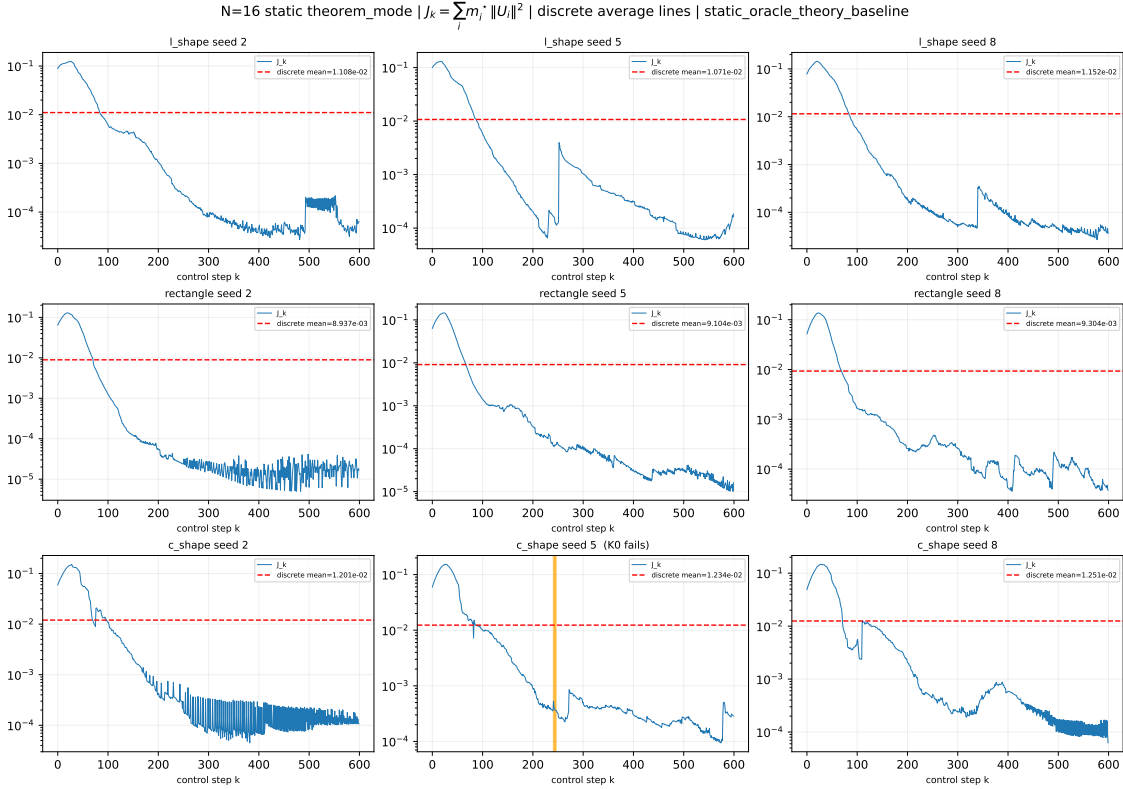


Figure 1: $J_k = \sum_i m_i^* \|U_i\|^2$ per case with discrete-average lines.

5 Suggested paper revisions

1. Split interfaces: continuous A5-style claims vs. sampled hold + abort.

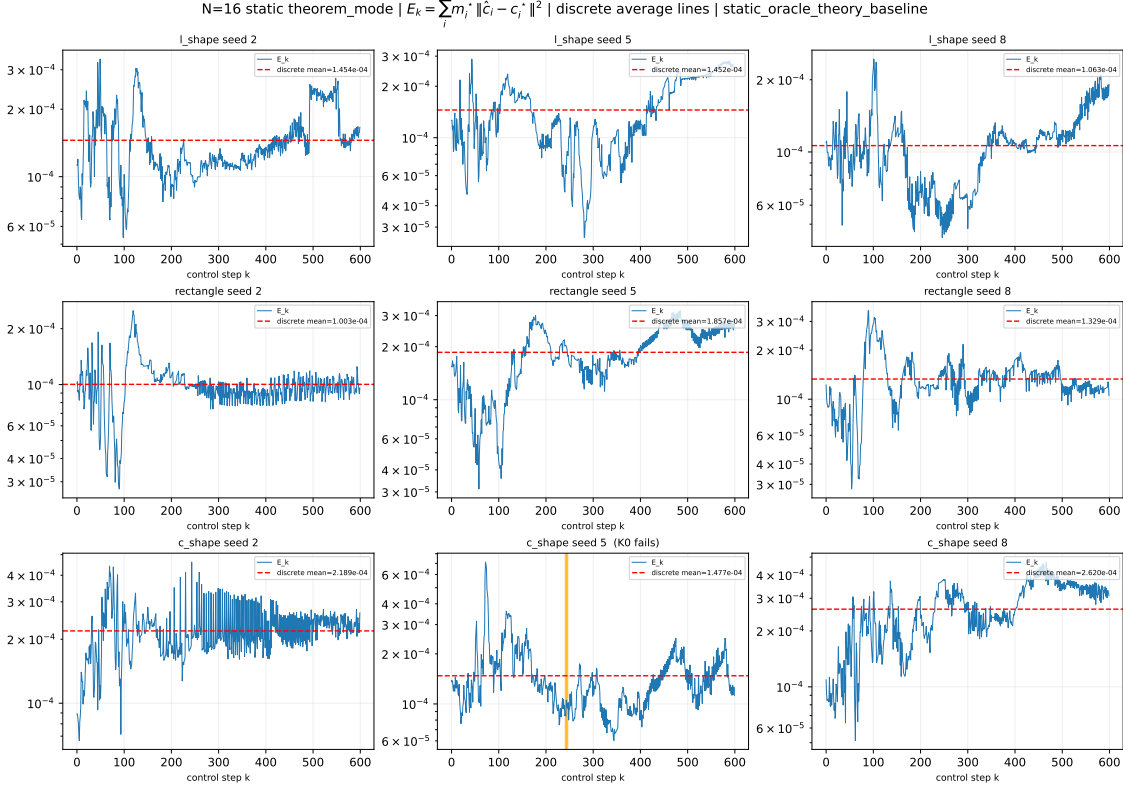


Figure 2: $E_k = \sum_i m_i^* \|\hat{c}_i - c_i^*\|^2$ (post-hoc) per case.

2. Retire the coarse Theorem 1 formula as the main quantitative claim; report geometry alongside any analytic bound.
3. Introduce the discrete candidate as Theorem 1' (sampled, fixed density) with Lemma A, $a > 0$, and explicit $\mathcal{K}_0$ conditioning.
4. Keep $e_i = \hat{c}_i - c_i^*$ and mass weights visible; separate J from $\|g^*\|^2$; mark post-hoc substitutions.
5. Write A3 with a single K_σ .
6. State the $N = 16$ Hessian as open.
7. Safety subsection: wall rows + no clip + abort; object hold via a Lipschitz clearance lower bound, not QP feasibility alone.

6 Open problems

1. A resolution-dependent a priori bound on $\|\hat{c}_{\text{grid}} - c^*\|$ tight enough that $(\star)$ beats Mu_{max}^2 without trajectory E : *not available*.
2. Closing dissipation for final QP commands outside $\mathcal{K}_0$ without deleting ρ .
3. $N = 16$ Hessian / conditional local-stability certificate.
4. Extending beyond static oracle to unknown moving objects with map-error bounds.

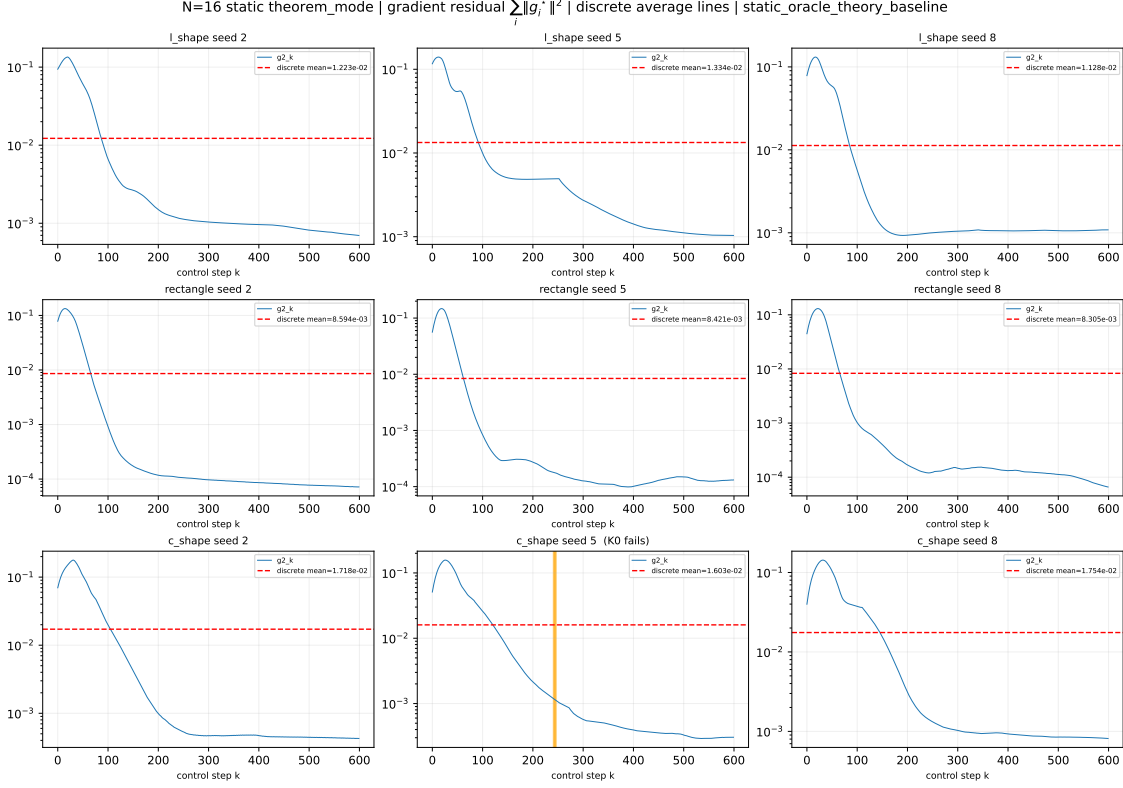


Figure 3: Gradient residual $\sum_i \|g_i^*\|^2$ per case (the original paper indicator; distinct from J).

5. Replacing hard-grid CVT if a continuous A5 statement is still desired.

A Corrected derivation (sampled, fixed density)

Throughout, sites are $P = (p_1, \dots, p_N)$ and commands $u = (u_1, \dots, u_N)$; the uncovered region is $\mathcal{U}(P) = D \setminus \bigcup_i \Omega_i(P)$ (we never reuse the symbol U for both the command and the uncovered set). The reference density $\phi^* \geq 0$ is *fixed* and the execution is sample-and-hold, $P_{k+1} = P_k + \Delta u_k$.

Status labels. *Proved* = mathematical under the stated hypotheses; *Conditional* = proved only on execution steps where named conditions hold, never inferred globally from finite runs; *Post-hoc numerical* = trajectory / quadrature evaluation; *Strict numerical certificate* = **not claimed** here (refined slacks remain $O(10^{-6})$ – $O(10^{-7})$, quadrature-limited).

A.1 Lemma A (frozen truncated-partition majorant)

Fix ϕ^* on D and radius $R > 0$. With truncated cells $\Omega_i(P) = V_i(P) \cap B(p_i, R)$ and

$$H^*(P) = \sum_i \int_{\Omega_i(P)} \|q - p_i\|^2 \phi^* dq + R^2 \int_{\mathcal{U}(P)} \phi^* dq,$$

define per-cell mass, centroid and (observer-compatible) gradient

$$m_i^* = \int_{\Omega_i} \phi^*, \quad c_i^* = \frac{1}{m_i^*} \int_{\Omega_i} q \phi^*, \quad g_i^* = 2 m_i^* (p_i - c_i^*).$$

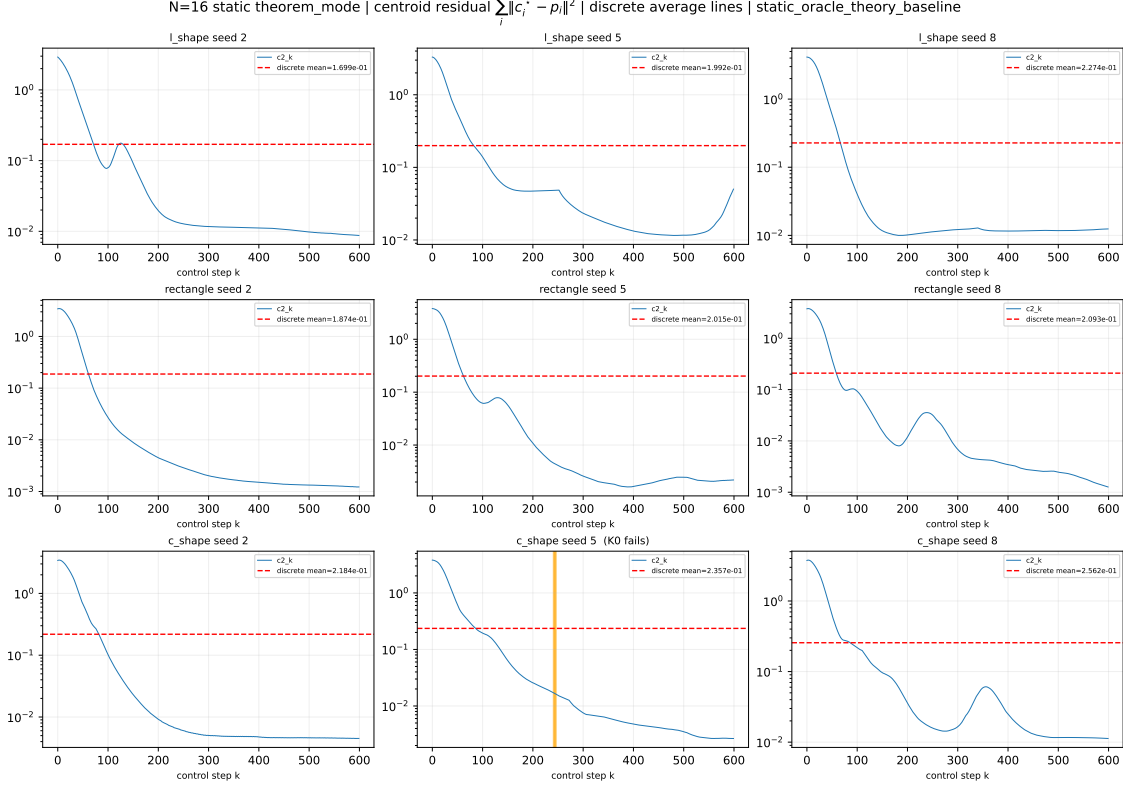


Figure 4: Centroid residual $\sum_i \|c_i^* - p_i\|^2$ per case.

At step k freeze the partition $\Omega_{i,k} = \Omega_i(P_k)$, $\mathcal{U}_k = \mathcal{U}(P_k)$ and set, for trial sites Q ,

$$\hat{H}_k(Q) = \sum_i \int_{\Omega_{i,k}} \|q - q_i\|^2 \phi^* + R^2 \int_{\mathcal{U}_k} \phi^*.$$

Then (1) $\hat{H}_k(P_k) = H^*(P_k)$; (2) $H^*(Q) \leq \hat{H}_k(Q)$ for every Q (re-partitioning can only lower the cost; the uncovered term is a Q -independent constant); (3) with $Q = P_k + \Delta u$,

$$\hat{H}_k(P_k + \Delta u) - \hat{H}_k(P_k) = \Delta g_k^{\star\top} u + \Delta^2 \sum_i m_{i,k}^* \|u_i\|^2.$$

Corollary (discrete inequality — proved).

$$H^*(P_k + \Delta u) - H^*(P_k) \leq \Delta g_k^{\star\top} u + \Delta^2 \sum_i m_{i,k}^* \|u_i\|^2. \quad (1)$$

Dissipation of H^* follows *when* the right-hand side is nonpositive; it is a consequence, not an assumption.

A.2 Sampling-period condition

Let $k_c > 0$ be the CVT gain and $\Delta > 0$ the hold. Define

$$a := \frac{2}{k_c} - \Delta.$$

Standing requirement $a > 0$, i.e. $\Delta < 2/k_c$. For the paper defaults $k_c = 0.9$, $\Delta = 0.05$ one has $a = 2/0.9 - 0.05 \approx 2.172 > 0$.

A.3 Per-robot effective gain and $\mathcal{K}_0$

With $d_i = \hat{c}_i - p_i$ (controller grid centroid minus site) and Euclidean-ball saturation, the executed command is parallel to d_i with an effective gain

$$\lambda_i = \min\left(k_c, \frac{u_{\max}}{\|d_i\|}\right) \quad (\|d_i\| > \varepsilon), \quad \lambda_i = k_c \text{ (idle)}.$$

Since $\lambda_i \leq k_c$, the per-robot slack $a_i = 2/\lambda_i - \Delta \geq a > 0$; hence $a_{\text{eff}} = \min_i a_i \geq a$ and the sampling condition never fails because of saturation. Let $F_{\rho,i,k}$ be the *final* planar QP feasible set (agent + wall + object rows including the ISSf margin ρ , and $\|u\| \leq u_{\max}$). Define the qualifying step set

$$\mathcal{K}_0 := \{k : \text{status}_i = \text{optimal and } 0 \in F_{\rho,i,k} \ \forall i, \text{ and } a > 0\}.$$

The margin-free flag `zero_input_feasible` is *not* sufficient; the predicate used is `zero_input_feasible_with_rho`. We do **not** promote “ $0 \in F_\rho$ on some L-runs” to a global hypothesis.

A.4 Theorem B (exact-centroid dissipation on $\mathcal{K}_0$)

Under Lemma A, $a > 0$, exact centroids $\hat{c}_i = c_i^*$, CVT $u_i^{\text{cvt}} = k_c(c_i^* - p_i)$, ball saturation $u_i^{\text{sat}} = \lambda_i(c_i^* - p_i)$ with the per-robot λ_i above, and final $u_i = \Pi_{F_{\rho,i,k}}(u_i^{\text{sat}})$ for $k \in \mathcal{K}_0$:

$$g_k^*{}^\top u^{\text{sat}} = - \sum_i \frac{2}{\lambda_i} m_i^* \|u_i^{\text{sat}}\|^2 \leq 0.$$

Projection optimality $(u^{\text{sat}} - u)^\top u \geq 0$ with $0 \in F_\rho$ then gives $(p_i - c_i^*)^\top u_i \leq -\|u_i\|^2/\lambda_i$ whenever $\lambda_i > 0$, hence

$$g_k^*{}^\top u \leq - \sum_i \frac{2}{\lambda_i} m_i^* \|u_i\|^2 \leq -\frac{2}{k_c} J_k$$

(using $\lambda_i \leq k_c$). Combined with (1) and $a > 0$,

$$H^*(P_{k+1}) - H^*(P_k) \leq -a\Delta J_k \quad (k \in \mathcal{K}_0).$$

Not claimed: exact centroids in the implemented grid controller; a global $\mathcal{K}_0 = [0, K)$; a continuous-time Theorem 1 replacement.

A.5 Theorem C (approximate centroids, saturation, final QP — finite-horizon J)

Let $e_i = \hat{c}_i - c_i^*$ and

$$J_k = \sum_i m_{i,k}^* \|u_{i,k}\|^2, \quad E_k = \sum_i m_{i,k}^* \|e_{i,k}\|^2,$$

with $\bar{J} = \frac{1}{K} \sum_k J_k$, $\bar{E} = \frac{1}{K} \sum_k E_k$. Assume the *entire* summation window satisfies $\{0, \dots, K-1\} \subseteq \mathcal{K}_0$, $a > 0$, and the implemented cascade

$$u_i^{\text{cvt}} = k_c(\hat{c}_i - p_i), \quad u_i^{\text{sat}} = \lambda_i(\hat{c}_i - p_i), \quad u_i = \Pi_{F_{\rho,i,k}}(u_i^{\text{sat}}).$$

Write $g_i^* = 2m_i^*(p_i - c_i^*) = 2m_i^*(p_i - \hat{c}_i) + 2m_i^*e_i$. Projection optimality against $0 \in F_{\rho,i,k}$ yields $u_i^\top u_i^{\text{sat}} \geq \|u_i\|^2$, hence $(p_i - \hat{c}_i)^\top u_i \leq -\|u_i\|^2/\lambda_i$ when $\lambda_i > 0$. Therefore

$$\begin{aligned} g_k^{*\top} u &= \sum_i 2m_i^*(p_i - \hat{c}_i)^\top u_i + \sum_i 2m_i^*e_i^\top u_i \\ &\leq -\sum_i \frac{2}{\lambda_i} m_i^* \|u_i\|^2 + 2 \sum_i m_i^* \|e_i\| \|u_i\| \\ &\leq -\frac{2}{k_c} J_k + 2\sqrt{E_k J_k}, \end{aligned}$$

where the last step uses $\lambda_i \leq k_c$ and Cauchy–Schwarz on the error sum. Lemma A then gives

$$H^*(P_{k+1}) - H^*(P_k) \leq -a\Delta J_k + 2\Delta\sqrt{E_k J_k}.$$

Young’s inequality $2\sqrt{EJ} \leq \frac{a}{2}J + \frac{2}{a}E$ absorbs the cross term:

$$H^*(P_{k+1}) - H^*(P_k) \leq -\frac{a\Delta}{2} J_k + \frac{2\Delta}{a} E_k.$$

Summing over $k = 0, \dots, K-1$ and using $H^* \geq 0$ produces the finite-horizon bound

$$\bar{J} \leq \frac{2H^*(P_0)}{aK\Delta} + \frac{4\bar{E}}{a^2}. \quad (\star)$$

Use of $(\star)$	Status
Identity under sat.+QP cascade, $\{0, \dots, K-1\} \subseteq \mathcal{K}_0$, Lemma A	Proved ($a > 0$)
Substituting trajectory $\bar{E}$ into $(\star)$	Post-hoc
Applying $(\star)$ when any step leaves $\mathcal{K}_0$	Inapplicable (do not stitch)
A priori $\bar{E}$ tight enough to beat $M_{\text{tot}}u_{\text{max}}^2$	Negative

J is not the gradient residual. $\|u\| \rightarrow 0$ does not imply $\|g^*\| \rightarrow 0$; J is mass-weighted command energy.

A.6 C-channel seed 5: inapplicable full-horizon bound

On the non-convex C-channel, seed 5, six agent-steps (frames 241–246, agent 14) violate $0 \in F_\rho$. Hence $\mathcal{K}_0 \neq [0, K)$, and $(\star)$ for the full horizon is **inapplicable**: we set `theorem_conditions_hold_full_horizon=full_horizon_J_bound_status="inapplicable"`, and record the post-hoc $\bar{J}, \bar{E}$ only as descriptive statistics tagged `not_covered_by_theorem`. We do **not** delete the failing steps to stitch a partial proof.

A.7 Negative a priori conclusion

Against the preferred geometric J ceiling $M_{\text{tot}}u_{\text{max}}^2$ one would need $\bar{E} \leq E^*$. The only model-derivable centroid-error bound is the reach-diameter bound $E \leq 4R^2M_{\text{tot}}$, which is *resolution-independent* (it certifies nothing about the 20×20 grid / oracle sampling accuracy) and numerically exceeds E^* . Therefore **no a priori centroid-error certificate** turns $(\star)$ into a quantitative improvement over geometry, and trajectory-measured E is explicitly *not* used as a uniform prior.

A.8 Three-command numerical check (consistency, not certificate)

For each stage $u \in \{u^{\text{cvt}}, u^{\text{sat}}, u^{\text{cmd}}\}$ we use the *hypothetical* hold map $P_k^+(u) = P_k + \Delta u$ and compare $\Delta H_{\text{hyp}}^*(u) = H^*(P_k^+) - H^*(P_k)$ with $\text{rhs}(u) = \Delta g_k^{*\top} u + \Delta^2 \sum m_{i,k}^* \|u_i\|^2$. On refinement, H , g and m are all recomputed on the *same* fine observer (the earlier code mixed a coarse-record m with fine H, g ; fixed here). Refined max slacks stay $O(10^{-6})$ – $O(10^{-7})$: numerical consistency at quadrature tolerance, **not** a strict numerical certificate (reason: `no_rigorous_quadrature_error_bound`).